

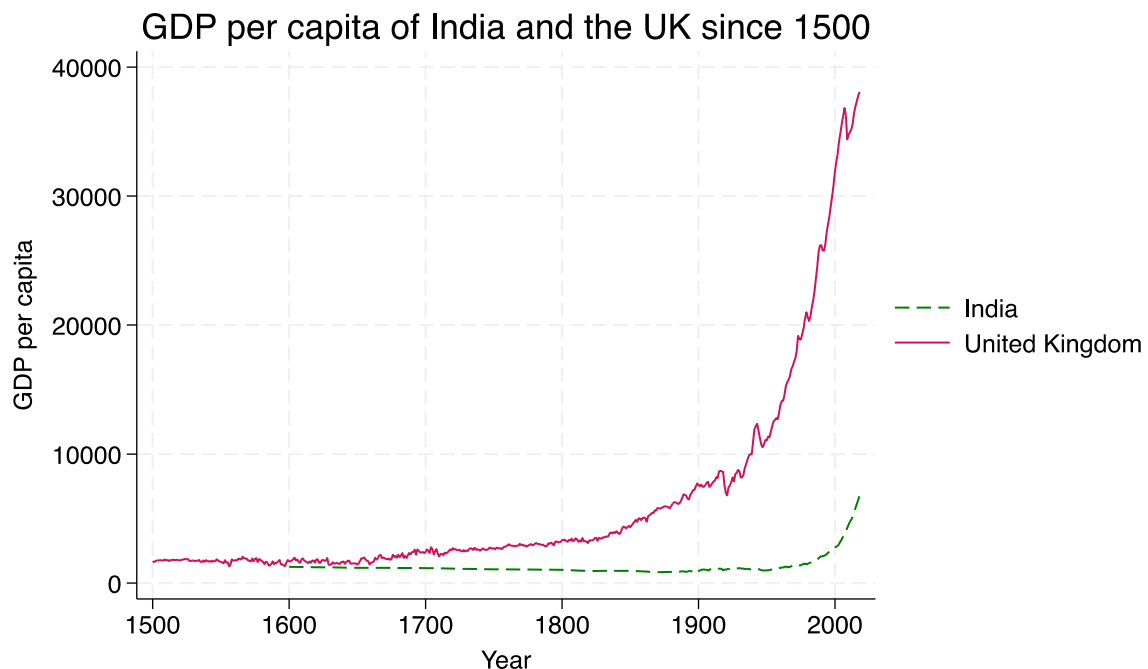
Political Economy of International Development

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Long run trends

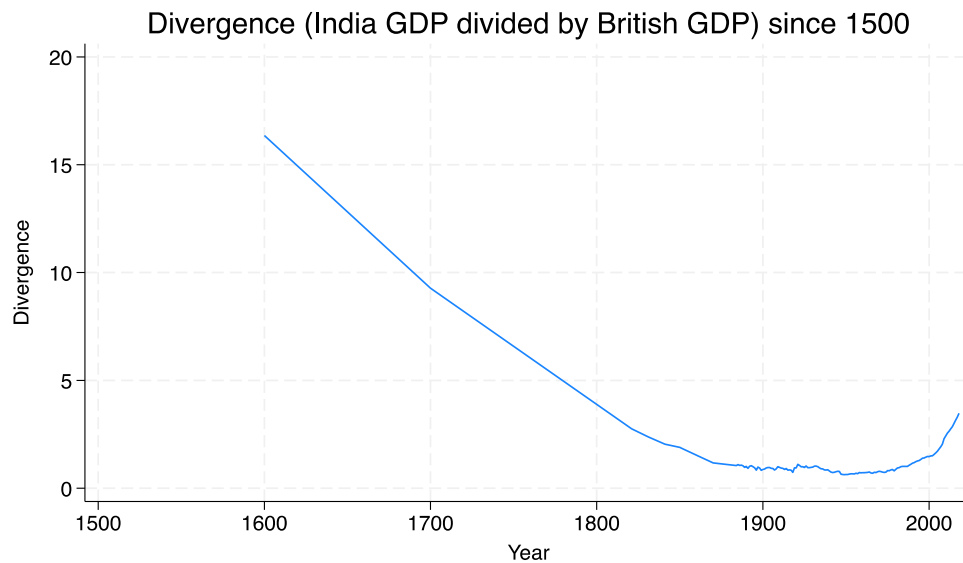
Line plot of GDP per capita for India and the UK since 1500 CE.

At around 1600, India's and the UK's GDP per capita was similar. However, since then it has fallen short. It has grown, however, not as fast as the UK's did. An upward trend in the Indian case is seen more prominently after independence in 1947, after mid-1900s.



Divergence

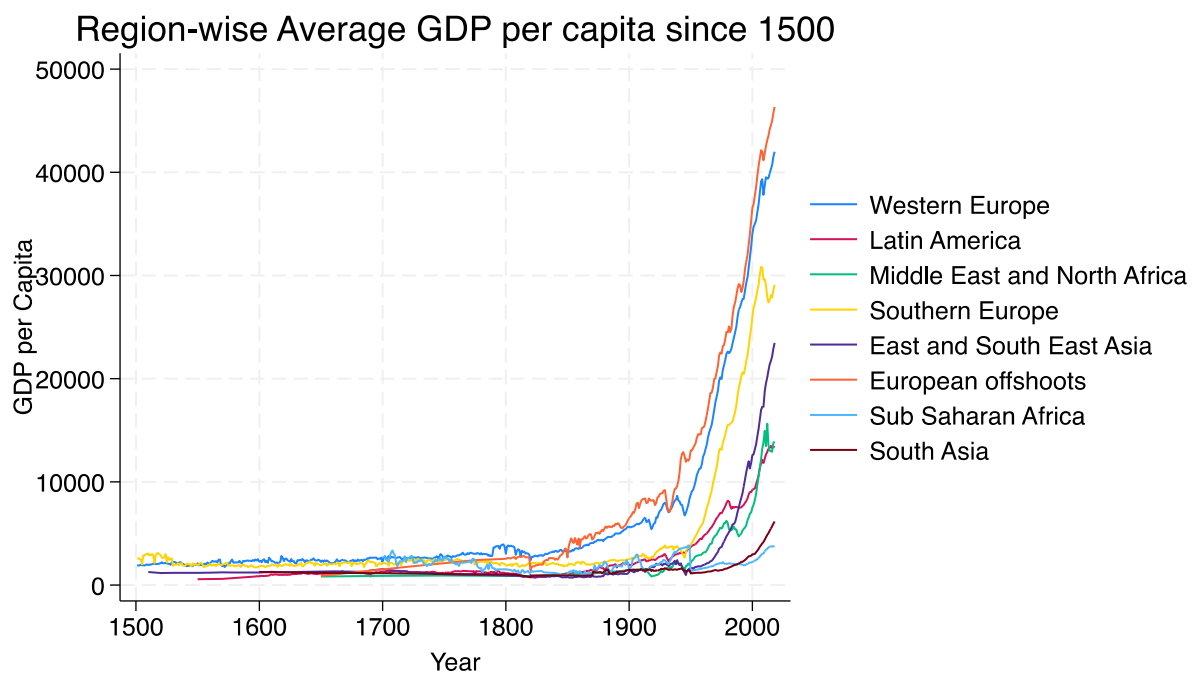
Divergence is defined as the ratio of India's GDP to that of the UK. The downward trend of this line from 1600 to mid-1900s shows that the UK's GDP was much larger compared to that of India, and the UK's GDP grew faster than India's. Hence, the ratio kept getting smaller and smaller. However, after the mid-1900s, a slight upward trend is seen. This means that India's GDP has started to grow relative to that of the UK. This can also be connected with the increase in Indian GDP per capita after the mid-1900s, as mentioned in the previous question.



Groups of countries to be made.

Groups namely, Western Europe (WE), Latin America (LA), Southern Europe (SE), East and South East Asia (ESEA), South Asia (SA), Sub-Saharan Africa (SSA), European offshoots (EO), and Middle East and North Africa (MENA) are made as mentioned. Other countries that were not required for the analysis at the moment (such as Denmark, Sweden, etc.) were dropped. All countries below the Saharan desert are taken as part of SSA.

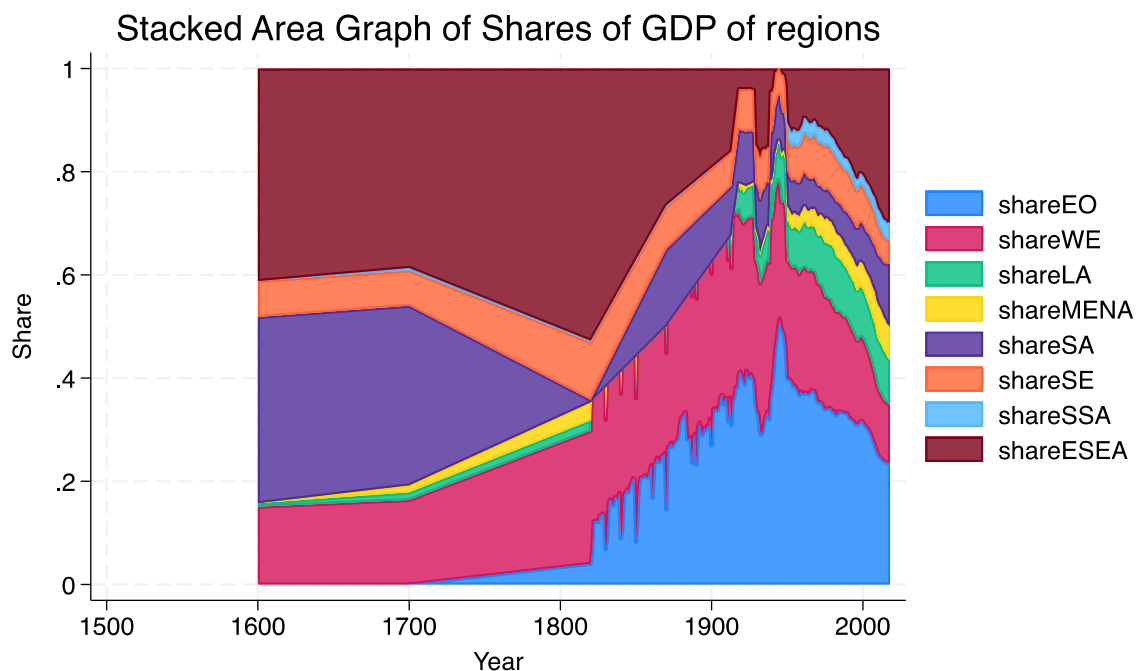
Average GDP per capita within each region over time since 1500 CE.



European offshoots and Western Europe are seen to be the frontrunners throughout since the 1500s. However, we can see that European offshoots overtook Western Europe (primarily, the colonizers) after 1800. This was also around the time when the US got its independence. As expected, Sub-Saharan Africa is the lowest around the year 2000 and has been low throughout. The upward trend in Western Europe, Southern Europe (to some extent), and European offshoots begins much before the upward trend for other regions. This can also indicate the

differences between the “developed” world and the “developing” world. The great divergence debate has seen many viewpoints – one being that the divergence was from the Middle Ages. However, scholars such as Pomeranz propose that it was later. Considering the obtained graphs, divergence or changes in shares of GDP are seen much after 1500, and not around the Middle Ages (this is also seen in the area graph shown in the next question).

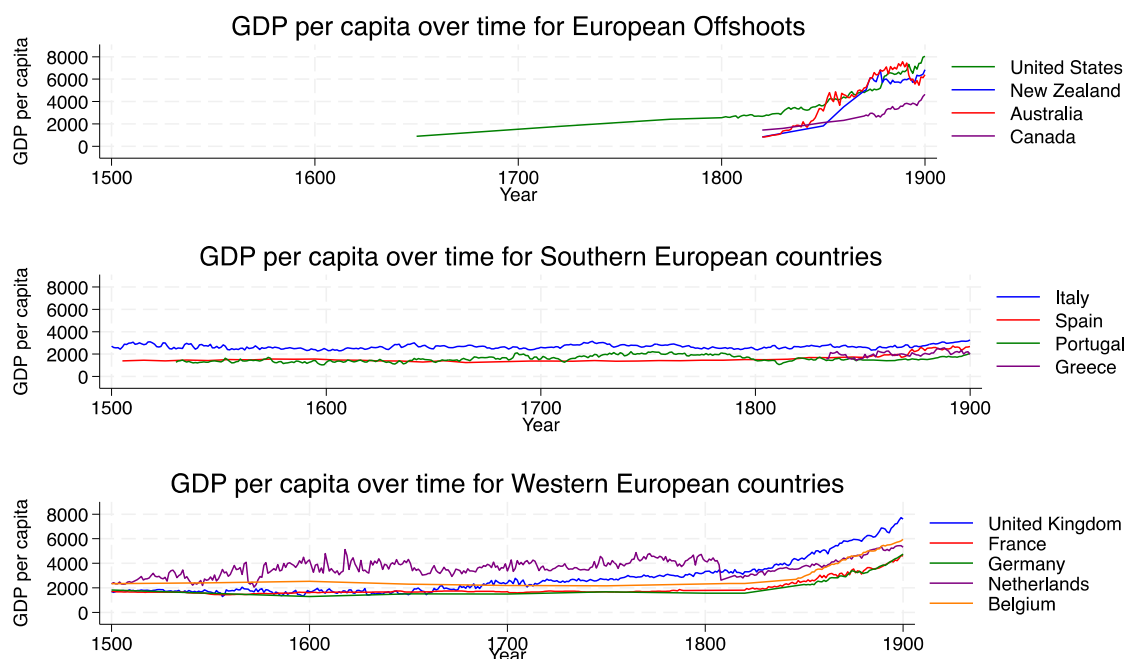
Share of world GDP accounted by each region over time



As it is seen, European offshoots and Western Europe again dominate the share of world GDP since 1800s. East and South East Asia share becomes lesser since then. South Asia is also seen to contribute quite a lot to world GDP before 1800s. This is also seen from the fact that India and China were booming centers of trade.

This reminds me of the video of the most populous cities that was shown in class. The most populous cities earlier were in South Asia or East Asia and Western Europe. However, over time they were overtaken by cities in European offshoot regions etc. This can perhaps be related to the contribution of these regions to the world GDP as well. Latin America, Sub Saharan Africa have less contribution to world GDP throughout.

GDP per capita in Western Europe, Southern Europe and European offshoots over time from 1500-1900.



The sharp upward trend is seen more prominently in the European offshoots, whereas the Southern European countries have more or less a relatively stable GDP per capita over time. Sharp growth is not seen. Unlike for Southern Europe, Western European countries (the original colonizers) have also had a relatively stable GDP per capita trend till 1800. Nonetheless, after 1800, their GDP per capita has seen a growth – just like the European offshoots did. This time period when Western Europe started seeing growth coincides with the industrial revolution. As expected, since the revolution began in the UK, Britain is the frontrunner in GDP per capita since 1800 in Western Europe. Some European offshoots such as Australia etc. became independent after 1800, and have grown since independence.

Estimating the impact of colonialism on today's GDP per capita.

The combined regression output for the whole exercise is given on the next page. Belgium is the base category for categorical variable 'colonizer'.

Regress log GDP per capita for 2018 on the variable colonizer

$$\ln(gdppc) = \beta_0 + \beta_1 colonizer + \epsilon$$

Interpretation: Here, the percentage change in GDP associated with the colonizer if the colonizer changes from Belgium to Britain is approximately 170% on an average. (Note this is a rough percentage calculation, have not used $e^\wedge$ in the precise calculation of the exact percentage change). Such changes in GDP per capita are higher for Italy, Portugal, Netherlands and Spain. Spain's coefficient is significant at the 0.01 level of significance as well. Spain was a big colonizer, especially in Latin America.

Regress GDP gap for 2018 on the variable colonizer

$$gdpgap = \beta_0 + \beta_1 colonizer + \epsilon$$

Interpretation: The change in colonizer from Belgium as base to Italy is associated with a 0.41 units increase in GDP gap on an average. Overall, the percentage changes in GDP gap are lesser than those for $\ln(\text{gdppc})$.

Regress log GDP per capita for 2018 on colonizer and colonization duration

$$\ln(\text{gdppc}) = \beta_0 + \beta_1 \text{colonizer} + \beta_2 \text{col_duration} + \epsilon$$

Interpretation: Keeping colonizer at base category, a one-year increase in colonization duration is associated with a 0.09% increase in GDP per capita on an average. (Again, approximate percentage, not the $e^{\wedge}$ one).

Regress GDP gap for 2018 on the variable colonizer and colonization duration

$$\text{gdpgap} = \beta_0 + \beta_1 \text{colonizer} + \beta_2 \text{col_duration} + \epsilon$$

Interpretation: Keeping colonizer at base category, a one-year increase in colonization duration is associated with a very small decline in GDP gap on an average.

Overall, these four regressions seem to indicate that colonization duration overall is negatively associated with GDP gap (favorable for colonies). Moreover, colonizers have positive coefficients, meaning, that they are positively related to $\ln(\text{gdppc})$ (again a bit favorable for colonies). However, we mustn't forget that colonization in itself is also contributed to the great divide between the global "North" and global "South".

Regression table on next page

Combined regression table

	(1)	(2)	(3)	(4)
Base: Belgium	ln gdppc	gdpgap	ln gdppc	gdpgap
Britain	1.707 (0.064)	0.175 (0.412)	1.696 (0.068)	0.177 (0.412)
France	1.346 (0.145)	0.121 (0.574)	1.352 (0.146)	0.120 (0.580)
Germany	0.581 (0.568)	0.0179 (0.940)	0.630 (0.541)	0.0114 (0.963)
Italy	2.860* (0.025)	0.415 (0.160)	2.903* (0.024)	0.410 (0.170)
Netherlands	2.637* (0.018)	0.231 (0.363)	2.475* (0.031)	0.253 (0.341)
Portugal	2.141* (0.028)	0.371 (0.101)	2.009* (0.046)	0.389 (0.098)
Spain	2.600** (0.006)	0.384 (0.077)	2.418* (0.015)	0.409 (0.078)
col duration			0.000946 (0.566)	-0.000126 (0.745)
cons	6.756*** (0.000)	0.0216 (0.917)	6.685*** (0.000)	0.0311 (0.883)
<i>N</i>	58	58	58	58
<i>R</i> ²	0.376	0.299	0.380	0.300
adj. <i>R</i> ²	0.288	0.201	0.279	0.186

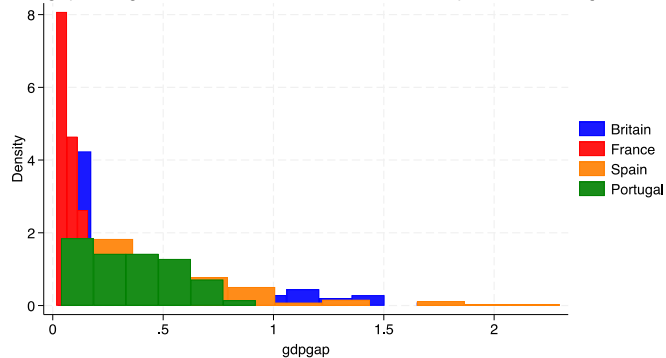
P values in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Histograms

GDP gap and colonizer histogram

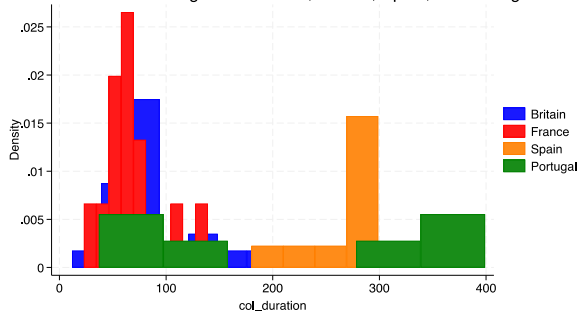
GDP gap histogram for colonizers Britain, France, Spain and Portugal



For France majorly, the GDP gap of colonies is less than one. Overall for all colonizers, the distribution is positively skewed, with major frequency concentrated where GDP gap is very much less than one. Which means that for all colonizers, their colonies have very less GDP compared to colonizers themselves.

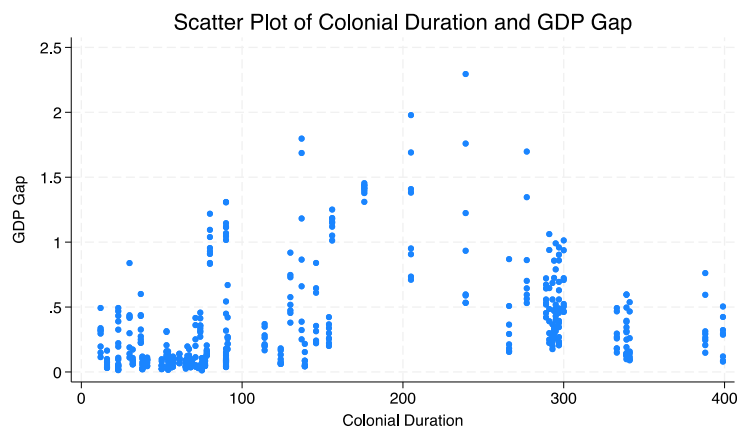
Duration of colonialism and colonizer histogram

Colonial Duration Histogram for Britain, France, Spain, and Portugal



Overall, France's colonial rule was mostly around 50-75 years (mode of the distribution). Similar for Britain. However, Britain did rule some colonies for up to 200 years. However, for Portugal there must be some colonies which were ruled for over 300 years. Spain is in between. The most number of years Spain ruled its colonies (especially Latin America) was around 275-300 years.

Scatter plot of duration of colonialism on the X axis and the GDP gap in the Y axis



The GDP gap for colonies which were under colonial rule for 0-150 years is very much less than one. Similarly, GDP gap for colonies which were under colonial rule for maximum number of centuries (300-400 years) also have a very less ($\text{GDP}_{\text{colony}}/\text{GDP}_{\text{colonizer}}$) ratio. This means that for these countries, their GDP relative to their colonizer's GDP is very less. Many of the Caribbean countries were colonized for up to 300 years. For instance Jamaica was under some or the other colonial rule from 1655 to 1962. Other such examples of high duration of colonization are India, Angola, Philippines etc. Countries which were under the colonial rule for a very short amount of time were Burundi, Rwanda, El Salvador etc. which also have a lesser GDP as of 2018. However, the United States was under colonial rule for around 170 years (from 1607 to 1776). This corresponds to the GDP gap greater than 1. This means that the GDP of the colonies is now greater than that of the colonizer. Even Australia was colonized for about 120 years. Similar trend as that of the US.